

SQL Notes

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1 About the notes

These notes cover the basics of SQL using the Northwind database in SQLite. This document is intended to provide a quick reference for SQL queries and operations, and is used in my 1st year information technologies course at Istanbul University. The notes include examples of SELECT, UPDATE, DELETE, and INSERT INTO statements, as well as the use of the LIMIT clause and various filtering techniques. The notes also include limited examples of JOIN operations and summary statistics using some aggregate functions (e.g., COUNT, SUM, AVG).

2 The Database Schema

The database is `Northwind.sqlite`, which is available to download at <https://github.com/jbytecode/sampledbs/tree/main/sqlite>. This database contains the following tables:

- Categories
- Customers
- Employees
- Order Details
- Orders
- Products
- Shippers
- Suppliers
- Regions
- Territories
- EmployeeTerritories

3 LIMIT

The `LIMIT` clause is used to specify the maximum number of records to return from a query. It is often used in conjunction with the `SELECT` statement to limit the number of rows returned.

4 SELECT

4.1 Basic Select

```
SELECT CompanyName, ContactName, City FROM Customers LIMIT 5;
```

CompanyName	ContactName	City
Alfreds Futterkiste	Maria Anders	Berlin
Ana Trujillo Emparedados y helados	Ana Trujillo	México D.F.
Antonio Moreno Taquería	Antonio Moreno	México D.F.
Around the Horn	Thomas Hardy	London
Berglunds snabbköp	Christina Berglund	Luleå

4.2 Filter by City

```
SELECT CompanyName, ContactName, City FROM Customers
WHERE City = 'Berlin'
LIMIT 5;
```

CompanyName	ContactName	City
Alfreds Futterkiste	Maria Anders	Berlin

4.3 Filter by Country and Order by Company Name

```
SELECT CompanyName, ContactName, City, Country FROM Customers
WHERE Country = 'UK'
ORDER BY CompanyName DESC
LIMIT 5;
```

CompanyName	ContactName	City	Country
Seven Seas Imports	Hari Kumar	London	UK
North/South	Simon Crowther	London	UK
Island Trading	Helen Bennett	Cowes	UK
Eastern Connection	Ann Devon	London	UK
Consolidated Holdings	Elizabeth Brown	London	UK

4.4 Filter by Country and Order by City and Country

```
SELECT CompanyName, ContactName, City, Country FROM Customers
WHERE Country IN ('UK', 'Germany')
ORDER BY Country ASC, City ASC
LIMIT 15;
```

CompanyName	ContactName	City	Country
Drachenblut Delikatessen	Sven Ottlieb	Aachen	Germany
Alfreds Futterkiste	Maria Anders	Berlin	Germany
Königlich Essen	Philip Cramer	Brandenburg	Germany
QUICK-Stop	Horst Kloss	Cunewalde	Germany
Lehmanns Marktstand	Renate Messner	Frankfurt a.M.	Germany
Ottilies Käseladen	Henriette Pfalzheim	Köln	Germany
Morgenstern Gesundkost	Alexander Feuer	Leipzig	Germany
Blauer See Delikatessen	Hanna Moos	Mannheim	Germany
Frankenversand	Peter Franken	München	Germany
Toms Spezialitäten	Karin Josephs	Münster	Germany
Die Wandernde Kuh	Rita Müller	Stuttgart	Germany
Island Trading	Helen Bennett	Cowes	UK
Around the Horn	Thomas Hardy	London	UK
B's Beverages	Victoria Ashworth	London	UK
Consolidated Holdings	Elizabeth Brown	London	UK

4.5 Filter by Phone Number

```
SELECT CompanyName, Phone
FROM Customers
WHERE Phone LIKE '(91)%'
LIMIT 5;
```

CompanyName	Phone
Bólido Comidas preparadas	(91) 555 22 82
FISSA Fabrica Inter. Salchichas S.A.	(91) 555 94 44
Romero y tomillo	(91) 745 6200

4.6 Filter by Phone Number Ending with 82

```
SELECT CompanyName, Phone
FROM Customers
WHERE Phone LIKE '%82'
LIMIT 5;
```

CompanyName	Phone
Bólido Comidas preparadas	(91) 555 22 82
Consolidated Holdings	(171) 555-2282
Godos Cocina Típica	(95) 555 82 82
Gourmet Lanchonetes	(11) 555-9482

4.7 Filter by Phone Number Containing Hyphen

```
SELECT CompanyName, Phone
FROM Customers
WHERE Phone LIKE '%-%'
LIMIT 5;
```

CompanyName	Phone
Alfreds Futterkiste	030-0074321
Ana Trujillo Emparedados y helados	(5) 555-4729
Antonio Moreno Taquería	(5) 555-3932
Around the Horn	(171) 555-7788
Berglunds snabbköp	0921-12 34 65

4.8 Filter by Fax Number Null (Empty)

```
SELECT CompanyName, Fax
FROM Customers
WHERE Fax IS NULL
LIMIT 5;
```

CompanyName	Fax
Antonio Moreno Taquería	
B's Beverages	
Chop-suey Chinese	
Comércio Mineiro	
Familia Arquibaldo	

4.9 Filter by Fax Number Not Null (Not Empty)

```
SELECT CompanyName, Fax
FROM Customers
WHERE Fax IS NOT NULL
LIMIT 5;
```

CompanyName	Fax
Alfreds Futterkiste	030-0076545
Ana Trujillo Emparedados y helados	(5) 555-3745
Around the Horn	(171) 555-6750
Berglunds snabbköp	0921-12 34 67
Blauer See Delikatessen	0621-08924

4.10 Products in Stock Ordered by Price

```
SELECT ProductName, UnitPrice, UnitsInStock
FROM Products
WHERE
UnitsInStock > 0
ORDER BY UnitPrice ASC
LIMIT 5;
```

ProductName	UnitPrice	UnitsInStock
Geitost	2.5	112
Guaraná Fantástica	4.5	20
Konbu	6	24
Filo Mix	7	38
Tourtière	7.45	21

4.11 Products Out of Stock

```
SELECT ProductName, UnitPrice, UnitsInStock
FROM Products
WHERE
UnitsInStock = 0
LIMIT 5;
```

ProductName	UnitPrice	UnitsInStock
Chef Anton's Gumbo Mix	21.35	0
Alice Mutton	39	0
Thüringer Rostbratwurst	123.79	0
Gorgonzola Telino	12.5	0
Perth Pasties	32.8	0

4.12 Products and Their Suppliers

```
SELECT ProductName, SupplierID
FROM Products
LIMIT 5;
```

ProductName	SupplierID
Chai	1
Chang	1
Aniseed Syrup	1
Chef Anton's Cajun Seasoning	2
Chef Anton's Gumbo Mix	2

4.13 Suppliers with Specific IDs

```
SELECT SupplierID, CompanyName, ContactName
FROM Suppliers
WHERE
SupplierID IN (1, 2);
```

SupplierID	CompanyName	ContactName
1	Exotic Liquids	Charlotte Cooper
2	New Orleans Cajun Delights	Shelley Burke

4.14 Products and Their Suppliers - Inner Join

```
SELECT Products.ProductName, Suppliers.CompanyName
FROM Products
INNER JOIN Suppliers
ON Products.SupplierID = Suppliers.SupplierID
LIMIT 5;
```

ProductName	CompanyName
Chai	Exotic Liquids
Chang	Exotic Liquids
Aniseed Syrup	Exotic Liquids
Chef Anton's Cajun Seasoning	New Orleans Cajun Delights
Chef Anton's Gumbo Mix	New Orleans Cajun Delights

4.15 Products with Unit Price Between 10 and 15

```
SELECT ProductID, ProductName, UnitPrice
FROM Products
WHERE
UnitPrice > 10 AND UnitPrice < 15
ORDER BY UnitPrice
LIMIT 5;
```

ProductID	ProductName	UnitPrice
46	Spegesild	12
31	Gorgonzola Telino	12.5
68	Scottish Longbreads	12.5
48	Chocolade	12.75
77	Original Frankfurter grüne Soße	13

4.16 OR

```
SELECT CompanyName, City from Customers
WHERE
City = 'Madrid' OR City = 'Portland'
ORDER BY City
LIMIT 5;
```

CompanyName	City
Bólido Comidas preparadas	Madrid
FISSA Fabrica Inter. Salchichas S.A.	Madrid
Romero y tomillo	Madrid
Lonesome Pine Restaurant	Portland
The Big Cheese	Portland

4.17 Summary Statistics

```
SELECT * FROM OrderDetails
WHERE
OrderID = 10248;
```

OrderID	ProductID	UnitPrice	Quantity	Discount
10248	11	14	12	0.0
10248	42	9.8	10	0.0
10248	72	34.8	5	0.0

```
SELECT ProductID, SUM(Quantity)
FROM OrderDetails
WHERE
OrderID = 10248
```

ProductID	SUM(Quantity)
11	27

```
SELECT ProductID, SUM(Quantity)
FROM 'Order Details'
```



```
GROUP BY ProductID
LIMIT 5;
```

ProductID	SUM(Quantity)
1	828
2	1057
3	328
4	453
5	298

4.18 Regions

```
SELECT * FROM Regions;
```

RegionID	RegionDescription
1	Eastern
2	Western
3	Northern
4	Southern

```
SELECT * FROM Regions
WHERE
RegionID < 3
ORDER BY RegionDescription DESC;
```

RegionID	RegionDescription
2	Western
1	Eastern

5 UPDATE

Description: Updates the unit price values of the products with product IDs 1, 2, 3, 4, and 5 by multiplying their current unit price values by 1.25.

5.1 Update Unit Prices of Specific Products

```
UPDATE Products
SET UnitPrice = UnitPrice * 1.25
WHERE ProductID IN (1, 2, 3, 4, 5);
```

5.2 Increase Units in Stock of Specific Products

Description: Increases the units in stock values of the products with product IDs 1 and 2 by adding 10 to their current units in stock values.

```
UPDATE Products
SET
UnitsInStock = UnitsInStock + 10
WHERE ProductID IN (1, 2);
```

5.3 Update Customer Information

Description: Updates the city and phone number of the customer with a customer ID of 1 to 'İstanbul' and '(90) 212 123 45 67', respectively.

```
UPDATE Customers
SET
City = 'İstanbul',
Phone = '(90) 212 123 45 67'
WHERE
CustomerID = 1;
```

6 DELETE

6.1 Delete Specific Products

Description: Deletes the products with product IDs 1, 2, 3, 4, and 5 from the Products table.

```
DELETE FROM Products
WHERE ProductID IN (1, 2, 3, 4, 5);
```

7 INSERT INTO

7.1 Insert a New Product

Description: Insert a new product with the name 'New Product', a supplier ID of 1, a category ID of 1, a unit price of 9.99, and 100 units in stock into the Products table.

```
INSERT INTO Products
(ProductName, SupplierID, CategoryID, UnitPrice, UnitsInStock)
VALUES
('New Product', 1, 1, 9.99, 100);
```

7.2 Insert a New Customer

Description: Insert a new customer with the name 'New Customer', a contact name of 'John Doe', a city of 'New York', and a country of 'USA' into the Customers table.

```
INSERT INTO Customers
(CustomerName, ContactName, City, Country)
VALUES
('New Customer', 'John Doe', 'New York', 'USA');
```

8 DROP TABLE

8.1 Delete Products Table

Description: Deletes the Products table from the database if it exists.

```
DROP TABLE IF EXISTS Products;
```